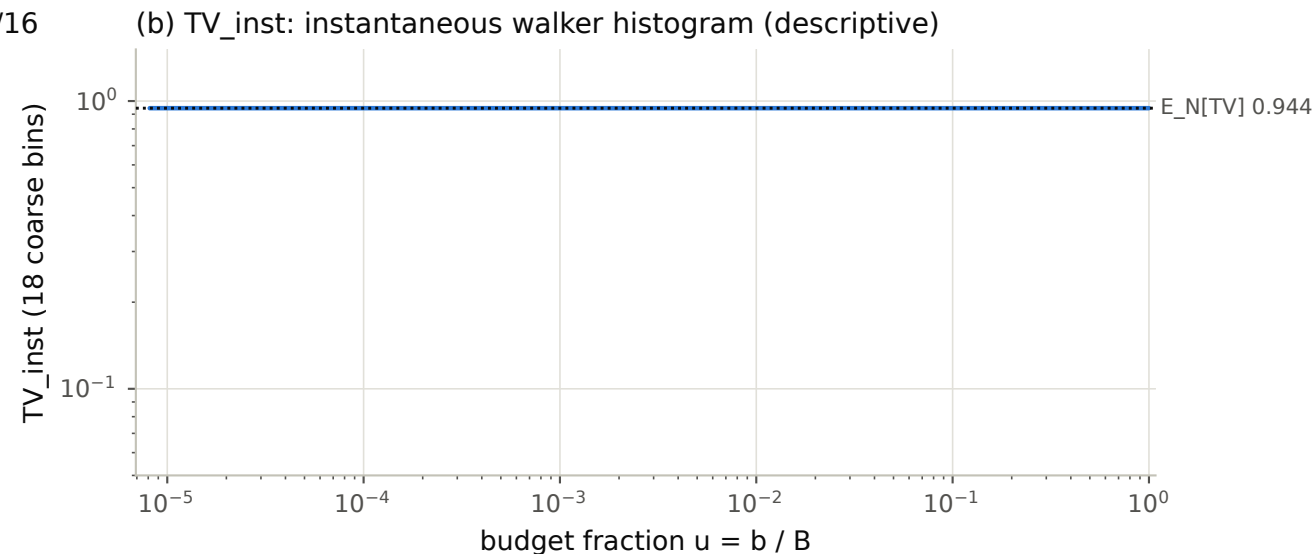
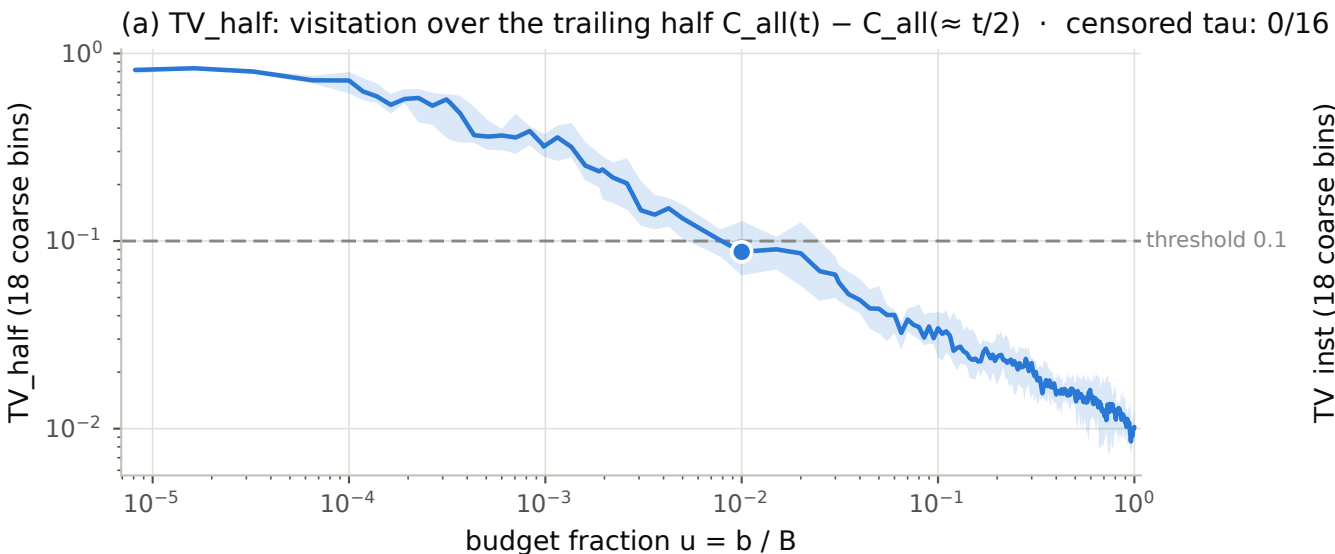


Ethane/LTA 300 K · N = 1 · T_N = 61440 t.u.

n_steps = 307,200,000 · h = 0.0002 · B = N n_steps = 3.072e+08 walker-steps · complete seeds: ABF 16/16 (N = 1: ABF only, no FR arm)

Marginal convergence: TV_half and TV_inst with the finite-N floor vs budget fraction u



- ABF (n = 16)
- O — TV_half threshold 0.1 (marker: persistent crossing of the median curve)
- interquartile range
- finite-N floor $E_N[TV] = 0.9444$ (exact, multinomial(N, uniform))

Line = median across seeds; band = interquartile range (25th-75th percentile) across seeds. Metrics are the frozen SCIENTIFIC_PLAN section-4 quantities, unsmoothed.

TV_half uses the saved step nearest $t/2$ (or the run start) as the window's lower edge (eqb_metrics.tv_half); tau for the marginal is defined on TV_half. TV_inst is never used for tau: at small N it is dominated by discreteness (17/18 at N = 1).

Small N (N = 1, i.e. ≤ 4): each seed contributes only 1 walker to an instantaneous histogram, so instantaneous marginals are necessarily discrete (TV_inst floor $E_N[TV] = 0.944$).

eqb_plot_config/2 · config lta_300K_production.json · 2026-10-10 07:12 UTC